

Gauss Elimination Method

High-Level Overview

The **Gauss Elimination Method** is a direct numerical technique used to solve systems of linear equations. It works by transforming the augmented matrix of a system into an **upper triangular form** (or row echelon form) using elementary row operations, after which the unknowns are easily determined via **back substitution**.

1. Principles and Workflow

- Direct Method:** Unlike iterative methods, it solves the system in a finite number of precise steps.
- Pivoting:** The leading diagonal elements are chosen as **pivot elements** (e.g., a_1, b'_2, c''_3).
 - Note on Leading Entries:** There is no strict restriction that the leading entry of a row must be 1 during the elimination process.
- Failure & Modification:** The method fails if any pivot element vanishes (0). This can typically be resolved by rearranging (swapping) the rows. If no non-zero pivot can be found by swapping, the matrix is singular and the system has no unique/standard solution.

2. Step-by-Step Procedure for a 3×3 System

Consider a general system of three linear equations:

$$\begin{aligned}a_1x + b_1y + c_1z &= d_1 \\a_2x + b_2y + c_2z &= d_2 \\a_3x + b_3y + c_3z &= d_3\end{aligned}$$

Step I: Construct the Augmented Matrix

Write the system in matrix form $[A \mid B]$:

$$[A \mid B] = \left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array} \right]$$

Step II: Eliminate x from Lower Rows

Assuming $a_1 \neq 0$, eliminate x from the second and third rows using row operations:

- $R_2 \rightarrow R_2 - \left(\frac{a_2}{a_1}\right) R_1$
- $R_3 \rightarrow R_3 - \left(\frac{a_3}{a_1}\right) R_1$

This yields an intermediate matrix where the first column has zeros below the main diagonal:

$$\left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ 0 & b'_2 & c'_2 & d'_2 \\ 0 & b'_3 & c'_3 & d'_3 \end{array} \right]$$

Step III: Eliminate y from the Third Row

Using the new pivot element b'_2 in the second row, eliminate y from the third row:

- $R_3 \rightarrow R_3 - \left(\frac{b'_3}{b'_2}\right) R_2$

This reduces the coefficient matrix to **upper triangular form**:

$$\left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ 0 & b'_2 & c'_2 & d'_2 \\ 0 & 0 & c''_3 & d''_3 \end{array} \right]$$

Step IV: Back Substitution

Translate the final upper triangular matrix back into equations:

- $c''_3 z = d''_3 \implies \mathbf{z} = \frac{d''_3}{c''_3}$
 - Substitute z into the second equation to solve for $\mathbf{y}$.
 - Substitute y and z into the first equation to solve for $\mathbf{x}$.
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3. Solved Examples

Example 1: Unique Solution (3×3)

Problem: Solve by Gauss elimination:

$$3x - 4y + 5z = -1, \quad -3x + 2y + z = 1, \quad 6x + 8y - z = 35$$

Solution:

1. **Augmented Matrix:**

$$[A \mid B] = \left[\begin{array}{ccc|c} 3 & -4 & 5 & -1 \\ -3 & 2 & 1 & 1 \\ 6 & 8 & -1 & 35 \end{array} \right]$$

2. **Row Operations:**

- Apply $R_2 \rightarrow R_2 + R_1$ and $R_3 \rightarrow R_3 - 2R_1$:

$$\sim \left[\begin{array}{ccc|c} 3 & -4 & 5 & -1 \\ 0 & -2 & 6 & 0 \\ 0 & 16 & -11 & 37 \end{array} \right]$$

- Apply $R_3 \rightarrow R_3 + 8R_2$:

$$\sim \left[\begin{array}{ccc|c} 3 & -4 & 5 & -1 \\ 0 & -2 & 6 & 0 \\ 0 & 0 & 37 & 37 \end{array} \right]$$

3. **Back Substitution:**

- $37z = 37 \implies \mathbf{z} = \mathbf{1}$
- $-2y + 6(1) = 0 \implies -2y = -6 \implies \mathbf{y} = \mathbf{3}$
- $3x - 4(3) + 5(1) = -1 \implies 3x - 7 = -1 \implies 3x = 6 \implies \mathbf{x} = \mathbf{2}$

Final Answer: $x = 2, y = 3, z = 1$.

Example 2: Infinitely Many Solutions

Problem: Test for consistency and solve:

$$x - z = 2, \quad y - z = -2, \quad x + y - 2z = 0$$

Solution:

1. **Augmented Matrix & Row Reductions:**

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & -1 & -2 \\ 1 & 1 & -2 & 0 \end{array} \right]$$

- Apply $R_3 \rightarrow R_3 - R_1$, then $R_3 \rightarrow R_3 - R_2$:

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

2. Rank Analysis:

- $\text{rank}(A) = \text{rank}(A \mid B) = 2$, which is less than the number of unknowns ($n = 3$).
- The system is **consistent and has infinitely many solutions**.

3. Parameter Assignment:

- Since $n - r = 3 - 2 = 1$, we can assign an arbitrary value to one variable. Let $z = 0$.
- $y - 0 = -2 \implies \mathbf{y} = -2$
- $x - 0 = 2 \implies \mathbf{x} = 2$

Final Answer: One particular solution is $(2, -2, 0)$.

Example 3: Systems with Four Unknowns (4×4)

Problem: Solve the system:

$$\begin{aligned} 5x + y + z + w &= 4 \\ x + 7y + z + w &= 12 \\ x + y + 6z + w &= -5 \\ x + y + z + 4w &= -6 \end{aligned}$$

Solution:

1. **Rearrange and Form Augmented Matrix:** Swap R_1 and R_4 to place a leading 1 at the top left for smoother calculation:

$$[A \mid B] = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 4 & -6 \\ 1 & 7 & 1 & 1 & 12 \\ 1 & 1 & 6 & 1 & -5 \\ 5 & 1 & 1 & 1 & 4 \end{array} \right]$$

2. Perform Row Operations:

- $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$, $R_4 \rightarrow R_4 - 5R_1$:

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 1 & 4 & -6 \\ 0 & 6 & 0 & -3 & 18 \\ 0 & 0 & 5 & -3 & 1 \\ 0 & -4 & -4 & -19 & 34 \end{array} \right]$$

- Further elimination steps yield the upper triangular form:

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 1 & 4 & -6 \\ 0 & 6 & 0 & -3 & 18 \\ 0 & 0 & 5 & -3 & 1 \\ 0 & 0 & 0 & -117 & 234 \end{array} \right]$$

3. Back Substitution:

- $-117w = 234 \implies \mathbf{w} = -2$
- $5z - 3(-2) = 1 \implies 5z + 6 = 1 \implies 5z = -5 \implies \mathbf{z} = -1$
- $6y - 3(-2) = 18 \implies 6y + 6 = 18 \implies 6y = 12 \implies \mathbf{y} = 2$
- $x + (2) + (-1) + 4(-2) = -6 \implies x - 7 = -6 \implies \mathbf{x} = 1$

Final Answer: $(x, y, z, w) = (1, 2, -1, -2)$.